

RS003 · Pygame Turret Game

Replit / Python — First graphics game — pygame, coordinates, loops.

For parents: these are the key English coding words your child meets in this course, week by week. You don't need to code — just read a word together and ask your child to explain it back in their own words. The tiny example shows what it looks like in real code.

W1

WORD	WHAT IT MEANS	TINY EXAMPLE
<code>pygame.init()</code>	wake up the pygame engine	<pre>pygame.init()</pre>
<code>display.set_mode</code>	open a game window of a given size	<pre>pygame.display.set_mode((800, 600))</pre>
<code>event loop</code>	the loop that checks for clicks and keys	<pre>for e in pygame.event.get():</pre>
<code>QUIT event</code>	the signal that the user closed the window	<pre>if e.type == pygame.QUIT:</pre>

W2

WORD	WHAT IT MEANS	TINY EXAMPLE
<code>surface</code>	any drawable image in pygame	<pre>screen = pygame.display.set_mode(...)</pre>
<code>blit</code>	paste one surface onto another	<pre>screen.blit(gun, (x, y))</pre>
<code>rect</code>	a rectangle holding x, y, width, height	<pre>r = pygame.Rect(0, 0, 50, 50)</pre>

W3

WORD	WHAT IT MEANS	TINY EXAMPLE
<code>def</code>	the keyword that defines a function	<pre>def draw_gun(): ...</pre>
<code>parameter</code>	an input the function expects	<pre>def draw_gun(x):</pre>
<code>call site</code>	the place where you use the function	<pre>draw_gun(100)</pre>

W4

WORD	WHAT IT MEANS	TINY EXAMPLE
<code>get_pressed</code>	which keys are held down right now	<pre>keys = pygame.key.get_pressed()</pre>
<code>K_LEFT / K_RIGHT</code>	key constants for the arrow keys	<pre>if keys[pygame.K_LEFT]:</pre>
<code>velocity</code>	how fast something moves per frame	<pre>x += velocity</pre>

W5

WORD	WHAT IT MEANS	TINY EXAMPLE
clamp	force a value to stay inside a range	<pre>x = max(0, min(x, 800))</pre>
min / max	smallest / largest allowed value	<pre>max(0, x)</pre>
boundary	the edge of the playable area	<pre>if x > screen_width: x = screen_width</pre>

W6

WORD	WHAT IT MEANS	TINY EXAMPLE
for loop	repeat once per item	<pre>for i in range(5): shoot()</pre>
range()	a sequence of numbers from start to end-1	<pre>range(3) # 0,1,2</pre>
iteration	one pass through the loop	<pre>for bullet in bullets:</pre>

W7

WORD	WHAT IT MEANS	TINY EXAMPLE
modular code	code split into small reusable pieces	<pre>def move(): ... def draw(): ...</pre>
single responsibility	one function does one thing	<pre>def fire_bullet(): ...</pre>
naming	good names describe the action	<pre>def reset_score(): ...</pre>

W8

WORD	WHAT IT MEANS	TINY EXAMPLE
score	the number that tracks player success	<pre>score += 1</pre>
game over	the state when the player has lost	<pre>if lives == 0: game_over = True</pre>
frame rate	how many frames are drawn per second	<pre>clock.tick(60)</pre>